

Language Learning in Virtual Reality: Enhancement of Long-Term Vocabulary Recognition and Understanding through Full-Body Avatars

Benjamin Lukas Schnitzer*
Affective and Cognitive Institute
Offenburg University
Offenburg, Germany
benjamin.schnitzer@hs-offenburg.de

Stefano Gampe*
Offenburg University
Offenburg, Germany
stefano.gampe@hs-offenburg.de

Arndt Bieberstein
Affective and Cognitive Institute
Offenburg University
Offenburg, Germany
arndt.bieberstein@hs-offenburg.de

Isabel Hoffmann
Ludwig-Maximilians-Universität
Munich, Germany
i.hoffmann@daf.lmu.de

Daniel Görlich
Media
Offenburg University
Offenburg, Germany
daniel.goerlich@hs-offenburg.de

Oliver Korn
Affective & Cognitive Institute
Offenburg University
Offenburg, Germany
oliver.korn@acm.org

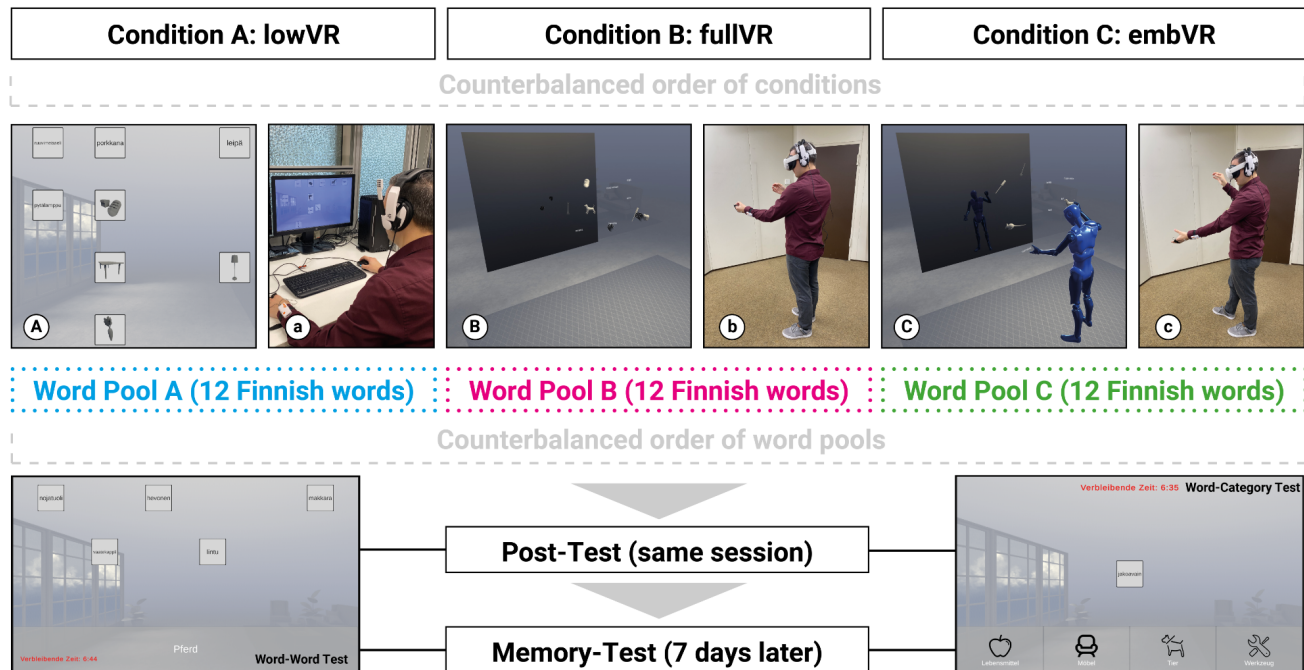


Figure 1: Within-participants study design with three conditions: (A) low immersive VR condition; (B) fully immersive VR condition with low embodiment; (C) fully immersive VR with high embodiment. Participants ($n = 32$) learned Finnish vocabulary by associating words and objects (either 2D images or 3D objects in fully immersive VR). Order of conditions and the order of the word pools (number of words per pool = 12) was counterbalanced. Learning outcomes were assessed through a post-test after the experimental learning phases and a memory test 7 days after the first session.

*Both authors contributed equally to this research.

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Abstract

Virtual Reality (VR) offers new embodied learning opportunities through full-body avatars interacting with the virtual environment and thereby inducing a sense of embodiment. Interestingly, while

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there is a profound body of research on various factors of embodiment as well as on embodied learning in virtual environments, the effects of virtual self-representation and avatar embodiment on learning outcomes are underexplored. In this work we present the results of an exploratory study ($n = 32$) based on a within-participants design with three conditions. We examined the effects of two fully immersive learning environments, one with low avatar embodiment and one with high avatar embodiment, and a low-immersive desktop application on vocabulary learning. The results imply that fully immersive VR does not improve short- and long-term recognition compared to desktop VR. However, increased immersion prevents knowledge loss and embodied interactions through full-body avatars can foster long-term vocabulary recognition and semantic understanding.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; **Virtual reality**; **HCI design and evaluation methods**.

Keywords

virtual reality, embodied language learning, avatar embodiment

ACM Reference Format:

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1 Introduction

Advances in fully immersive Virtual Reality (VR), predominantly experienced with head-mounted displays (HMDs), enable new embodied learning experiences [4] combining various input signals like hand-tracking [58], eye-tracking [7], physiological sensors [48], and haptic feedback [40]. Recent HMDs even provide tracking features for full-body avatars, without the need of additional and costly sensors [39]. Thus, apart from acquiring practical knowledge in specific simulations, today's learners can interact with and manipulate depictions of abstract concepts and make sense of them through multimodal sensorimotor stimulation [5]. An example is language acquisition: through spatial, kinesthetic, and playful interactions grammatical structures can be learned by virtually touching words and corresponding visuals. Such interactions are related to embodied cognition [56], a growing area of research in various disciplines like linguistics [24], education, and cognitive science [29].

Each discipline offers unique perspectives on embodiment but there are endeavors to conceptually unify the commonalities. One example is “4E cognition” [44], an umbrella term that encompasses different approaches. The underlying connection is the belief that intelligent behavior cannot be comprehended solely by examining internal brain processes. Accordingly, cognition is perceived as emerging from and being connected to interactions of an individual's physical body and its social environment.

Humans inevitably have an embodied experience of the world. Every learning experience with technology is consequently embodied, too. However, in contrast to low immersive VR such as desktop applications, the way we see or perceive our (virtual) body in fully immersive VR can be altered. Generally, full-body avatars allow users to experience a high sense of embodiment towards a virtual body [28] built on the illusion of body-ownership, agency, and self-location. However, the appearance of one's digital avatar can differ from the real-life body in terms of morphology, physiology, age, or gender [15, 34]. Moreover, the visual self-representation can be minimized to a non-embodiment or low-embodiment, e.g., by only showing the hands [61]. Johnson-Glenberg et al. [25] introduced a taxonomy by differentiating between four degrees of embodied learning. The degrees describe levels of sensory-motor engagement in learning environments, ranging from the fourth degree (high locomotion, immersive experience, and strong gestural mapping) to the first degree (seated, minimal movement, and low immersion) [26]. Offering a framework for embodied learning in VR, Chatain et al. identified three key approaches to embodied learning in fully immersive VR: (1) embodied cognition, emphasizing the role of bodily actions in learning; (2) embodied interaction, focusing on how the body aids in understanding interactions through meaning-making; and (3) avatar embodiment, linking the physical body with its digital representation [4].

There is empirical evidence for the positive impacts of immersive VR on foreign language learning [3, 47], embodied learning in immersive VR [5, 6, 43], and embodied language learning [21, 30, 42]. Moreover, there is a profound body of research on various factors such as appearance, agency, self-location, and ownership affecting the sense of embodiment [1, 9–11, 15, 16, 19, 28, 50, 51, 61]. At the same time, the effects of avatar embodiment and embodied interaction on language learning in fully immersive VR remain unclear. Considering the described framework in [4], we therefore examine two research questions in this work:

RQ1: Compared to low immersive VR, how does fully immersive VR affect foreign vocabulary learning in terms of short and long term recognition as well as semantic understanding?

RQ2: How does increased avatar embodiment and self-representation in fully immersive VR affect foreign vocabulary learning?

We hypothesize that, compared to desktop VR, fully immersive VR positively affects (**H1**) passive vocabulary recognition and (**H2**) semantic understanding due to multisensory input and engagement. Moreover, we hypothesize that increased avatar embodiment positively affects (**H3**) passive vocabulary recognition and (**H4**) semantic understanding. In the following section, we first describe related research. Then we report the results of a user study examining the stated hypotheses.

2 Related Work

A study by Kaplan-Rakowski et al. [27] with 371 participants showed that 2D images significantly outperformed stereoscopic 3D images in vocabulary recall tests. The authors attributed the poorer S3D performance to cognitive overload and distraction from immersive depth cues. While they did not include fully immersive VR or interaction (participants passively consumed content), their findings

suggest that increased immersion does not always enhance learning outcomes [27].

Another study by Papin and Kaplan-Rakowski [46] examined the impact of annotated 360° pictures on vocabulary learning across three conditions: high-immersion virtual reality, low-immersion virtual reality (360° pictures on a screen), and standard 2D pictures. Results of a study with 63 participants revealed that vocabulary learning with annotated 360° pictures viewed on a desktop monitor outperformed both the high immersive and the 2D condition in receptive vocabulary recall. The authors attributed the effectiveness to a balance of immersion and ease of use, as excessive stimuli of fully immersive VR may lead to cognitive overload [46].

At the same time, a comprehensive literature review by Dhimolea et al. [12] questions the previous results by highlighting that the perceptions of language learning in VR are positive, but its effectiveness is inconclusive. They further state that the “implications of VR technology for improving language learning outcomes need further investigation” [12, p. 11].

In “Words in Kitchen”, Jia and Liu [22] designed a virtual situated learning experience where users learn new vocabulary by interacting with virtual objects in a kitchen. A pilot study indicated that the system supports the recognition of words and stimulates interest based on the word-object association.

Similarly, Garcia et al. [18] designed a playful experience in fully immersive VR where users explore Spanish translations of everyday household items in a search-and-find format and score points for correctly identified objects. While the application generally provides situated language learning, there was no self-representation through a virtual body. In a preliminary study, users found the application to be more joyful than traditional learning, but no results regarding learning effectiveness were reported [18].

Hartfill et al. [20] investigated the effects of an immersive game called “Word Saber” which enables immersive learning of word-object associations through embodied interactions. While the game was found to be generally effective for vocabulary learning, scores in terms of recognition and recall were significantly lower compared to a non-digital flashcard method [20].

White et al. [59] developed a virtual classroom offering multisensory language learning via 2D/3D visualizations, audio, and interaction-based reinforcement. However, a direct word-object association through written language is not provided. Considering the importance of the meaning-making process in embodied cognition and embodied interactions, the application lacks distinct interactions that foster learning through bodily actions.

With “Words in Motion”, Vázquez et al. [57] presented a VR learning system that reinforces associative learning of word-action pairs by recognizing the learner’s movements and presenting the associated word in the target language. For example for the word wave, a wavy movement has to be performed with the help of the controllers. Results of a study suggest that the kinesthetic approach in immersive VR yields lower immediate learning gains than text-only methods and no difference from non-kinesthetic VR but results in significantly higher retention after a week [57].

In general, the described works did not account for avatar embodiment since self-representation through a virtual body was either not existent (no visual limbs) or very low (e.g., through a virtual lightsaber). The absence of self-representation through an avatar,

however, appears to impact learning. For example, Lan et al. [31] compared two types of embodied learning: real physical bodies and 3D avatars, with non-embodied learning. The study found that elementary students learning English as a foreign language performed better when watching their 3D avatars perform motions compared to physically enacting the motions or using non-embodied learning [31]. Moreover, these findings question the results that found no difference in learning outcomes with regards to increased immersion. Regarding avatar embodiment, a more recent study by Xia et al. [60] found higher recall in immediate and delayed post-interventions for adults learning language with the help of a VR system providing high embodiment compared to a group learning with a system providing low embodiment.

3 Methods

We conducted a study ($n = 32$) based on a within-participants design with three conditions and a word-object association scenario to investigate the stated research questions. This was done intentionally to compare the impact of different levels of immersion and embodiment on vocabulary learning while controlling for individual differences in cognitive abilities and learning strategies. Regarding **RQ1** and considering the reality-virtuality continuum [37], we differentiated between two extremes: a low immersive VR condition (*lowVR*) and a fully immersive VR (*fullVR*) condition. In the *lowVR* condition visual stimuli were presented on a regular 2D screen. The computer mouse functioned as an input system and auditory stimuli were provided through headphones (Fig. 1). This resembles traditional low immersive e-learning and game-based learning applications. In the *fullVR* condition, visual stimuli were provided by an HMD (Meta Quest 3) with hand and finger tracking. Users could use their own hands to interact with the virtual environment. Auditory feedback was provided through headphones plugged into the HMD. In contrast to the *lowVR* condition which did not provide any self-representation except the mouse cursor, users could see a virtual representation of their hands in the *fullVR* condition. Additionally, these hands were visible in a virtual mirror placed in front of the user (Fig. 1).

Regarding **RQ2**, we differentiated between the fully immersive VR (*fullVR*) and a fully immersive VR condition with high embodiment (*embVR*) in which users were represented by a gender-neutral full-body avatar. Similar to the *fullVR* condition, users could use their real hands to grasp the objects and move around within a marked playground by naturally walking around. The avatar made use of inverse kinematics features of the Meta SDK. Movements of the mannequin-like avatar were also visible in the virtual mirror in front of the user (Fig. 1). While they sat in the *lowVR* condition, participants were standing in an upright position in both fully immersive conditions.

To avoid prior knowledge, we chose the Finnish language rarely spoken by the population (German native speakers) and not being related to any frequently spoken language. We followed the methodology in [27]. Furthermore, every participant confirmed to have no prior knowledge of the language. Considering Ebbinghaus’ Forgetting Curve [13] and following the methodology of [27], [32], [52], and [45] an adequate number of target words appeared to be 36 resulting in three equally sized word pools. Hence, in the

learning phase participants learned a pool of 12 words per experimental condition by actively combining written Finnish words and corresponding visualisations and objects. This was achieved by matching the spatial position of words and objects, an embodied interaction that creates meaning through affiliation [4]. All visualizations were grayscale. While the visualisations in *lowVR* were 2D images and interaction was limited to 2D space, in both VR conditions these visuals were actual 3D objects with six dimensions of freedom. Participants were able to grab and move around these objects in 3D space.

We only selected concrete nouns of which each belongs to one of the four categories: (1) furniture, (2) food, (3) animals, and (4) tools. Thus each word pool consisted of three words for each category, also known as hypernyms. Since these hypernyms are semantically disjunct, we intentionally chose them for the post-learning assessment. Participants did not know about the categories in advance. Exemplary, Table 1 in the appendix shows word pool A.

We used the Unity 3D game engine to create the applications where we presented five words [38] and their corresponding visualization simultaneously. A pooling-algorithm randomly chose five words and five corresponding objects and placed them randomly on a grid-like matrix of positions. This was done to avoid serial position effects. The positions for the two fully immersive conditions were about an arm's length away. If a correct word-object pair was found both disappeared and auditory feedback was given: the Finnish word was read out by a native speaker, followed by the German translation. Additional visual feedback was provided through a colored frame [2]. In cases of an incorrectly matched pair, a red frame temporarily appeared around the screen in *lowVR*. Likewise a green frame appeared for a correctly matched pair. In both fully immersive conditions this frame was shown around the virtual mirror [2]. Subsequently, the pooling algorithm randomly selected a new word-object pair from the remaining seven items, placing it with the four previously chosen pairs on the grid. This process continued until fewer than five pairs remained, at which point only those pairs were shown. The entire word pool was presented three times per participant per condition. No time limit was set for these iterations. Conditions as well as word pools were counterbalanced to avoid biases and learning effects, resulting in 18 condition-pool permutations assigned sequentially to participants, cycling back to the first combination after p18.

Participants' memorization was assessed through a single-choice test conducted in the lab and consisting of two parts. In the first part, participants matched German target words to one of five Finnish options, relying on visuo-semantic associations and auditory feedback. The algorithm randomly chose the target word, its corresponding Finnish translation, and additional four foreign words until all target words had been presented. This *word-word test* examines associative memory and retention based on the image-foreign word link (Fig. 1). The second part required assigning Finnish words to hypernyms, testing semantic understanding and the ability to categorize. In this *word-category test* participants connected the foreign word to the four broader categories, such as linking the Finnish word for "dog" to "animals" (Fig. 1). This task generally involves deeper encoding and abstraction, with higher cognitive load and decision complexity. Both parts included 36 association tasks and a visible 6:30-minute countdown to simulate an exam-like scenario,

though all participants completed within this time. The word-word test as well as the word-category test were presented on a laptop and performed by using a mouse. The similar tests were repeated after seven days.

For this study, we solely employed quantitative measures. Learning outcome was measured as the absolute number of correctly matched pairs resulting in a total of 36 possible correct combinations per test and 12 correct combinations for each condition.

We computed an a-priori power analysis using G*Power 3.1. For repeated measures ANOVA (within factors) with a medium effect size ($f = .3$, α error prob = .05, power ($1 - \beta$ err prob) = .95, number of groups = 1, number of measurements = 3, corr among rep measures = .5, nonsphericity correction $\epsilon = 1$) the total sample size was 31. Hence, we recruited 32 volunteers (mean age = 22.59, SD = 4.75) of a convenience sample consisting of university students. 12 participants were female, 19 were male, and 1 was non-binary. All participants were 18 years or older and did not receive payment. After being informed about the procedure and data anonymity according to the Declaration of Helsinki and GDPR, participants were asked to give their written consent. Subsequently, participants filled in a demographic questionnaire. All participants conducted the study in two sessions in November and December 2024. In the first session, they first experienced all three experimental conditions at once, followed by the word-word and word-category post tests. In the second session seven days later, participants conducted the word-word and word-category memory tests (Fig. 1). As an undergraduate computer science class with a special focus on virtual worlds, the participating students had a comparatively high level of previous VR experience. Nevertheless, the calibration phase of the two VR conditions was used for a short familiarisation phase, where participants were verbally instructed on how to interact with the virtual objects. During this time, they could already see movements of their avatar or hands in the mirror in front of them. After both fully immersive VR conditions, we asked participants about their well-being to ensure that possible cybersickness symptoms did not bias the results.

4 Results

None of the participants reported cybersickness symptoms and all 32 data sets were included in the analysis. A maximum of 12 correct combinations was achievable for each condition among the four tests. Except for the results of the word-category memory test of the *fullVR* condition, Shapiro-Wilk tests indicated non-normally distributed data. Subsequent analysis with histograms underlined this assumption. Thus, only non-parametric tests using RStudio were applied. We used the packages stats, pgirmess for multiple comparisons after friedman test, and rcompanion to compute the statistics. Effect sizes for post-hoc Wilcoxon tests are reported as r calculated as Z divided by square root of the number of observations in one group. The interpretation of effect sizes is based on Cohen [8]. Due to differences in sensitivity between the Friedman test and paired Wilcoxon tests in small sample sizes, we also investigated differences with the help of pairwise comparisons, particularly when a global Friedman test did not indicate significant overall difference but its p values were close to the alpha level of .05. Figure 2 shows the boxplot results for both tests. Table 2 - 5 in the appendix

show the results for mean, standard deviation (SD), median, and interquartile range (IQR).

Word-Word Post Test: A Friedman test ($\chi^2 = 5.57$, $df = 2$, $p = .06$) indicated no significant differences among conditions. While not statistically significant, post-hoc multiple comparisons calculated with a critical difference Δ_c between conditions ($\alpha = .05$, $\Delta_c = 19.15$) showed bigger observable differences Δ_o between *lowVR* and *fullVR* ($\Delta_o = 13.50$) as well as *fullVR* and *embVR* ($\Delta_o = 16.50$) compared to *lowVR* and *embVR* ($\Delta_o = 3.00$). A paired Wilcoxon test with Bonferroni correction revealed a significant difference between the *fullVR* and *embVR* condition ($Z = -2.34$, $p < .05$, 95% CI $[-1.99, -0.00004]$, Hodges-Lehmann estimate $\Theta_{HL} = -1.00$) with a medium effect size ($r = .44$, 95% CI $[.10, .72]$).

Word-Word Memory Test: A Friedman test for the word-word memory test ($\chi^2 = 10.15$, $df = 2$, $p < .01$) yielded significant differences among conditions. Multiple post-hoc comparisons revealed a significant difference between the *fullVR* and the *embVR* condition ($\alpha = .05$, $\Delta_c = 19.15$, $\Delta_o = 23.5$). A paired post-hoc Wilcoxon test confirmed this difference ($Z = -2.28$, $p < .05$, 95% CI $[-2.50, -0.00004]$, $\Theta_{HL} = -1.50$) with a medium effect size ($r = .42$, 95% CI $[.09, .72]$).

Knowledge Loss for Word-Word Tests: With regards to knowledge loss, paired Wilcoxon tests using Bonferroni correction revealed a significant difference between the *lowVR* post test and the *lowVR* memory test ($Z = 2.64$, $p < .01$, 95% CI $[.50, 2.00]$, $\Theta_{HL} = 1.50$), with a medium effect size ($r = .55$, 95% CI $[-.82, -.21]$). Regarding the *fullVR* and *embVR* condition, paired Wilcoxon tests for the comparisons between post and memory tests were insignificant.

Word-Category Post Test: A Friedman test ($\chi^2 = 5.52$, $df = 2$, $p = .06$) did not indicate significant differences among conditions. However, multiple post-hoc comparisons between groups ($\alpha = .05$, $\Delta_c = 19.15$) revealed larger differences between *lowVR* and *fullVR* ($\Delta_o = 16.50$) and *lowVR* and *embVR* ($\Delta_o = 13.50$) than between *fullVR* and *embVR* ($\Delta_o = 3.00$). Subsequent pairwise Wilcoxon tests with Bonferroni correction yielded significant differences between *lowVR* and *fullVR* ($Z = 2.31$, $p < .05$, 95% CI $[0.000028, 2.77]$, $\Theta_{HL} = 1.50$) and *lowVR* and *embVR* ($Z = 2.21$, $p < .05$, 95% CI $[0.00003, 2.00]$, $\Theta_{HL} = 1.00$), both with medium effect sizes ($r = .47$, 95% CI $[.10, .78]$ and $r = .42$, 95% CI $[.06, .71]$, respectively).

Word-Category Memory Test: A Friedman test ($\chi^2 = 5.96$, $df = 2$, $p = .05$) did not yield any significant differences among conditions. Post-hoc comparisons ($\alpha = .05$, $\Delta_c = 19.15$) revealed larger differences between *lowVR* and *fullVR* ($\Delta_o = 17.00$) and *fullVR* and *embVR* ($\Delta_o = 14.50$), compared to *lowVR* and *embVR* ($\Delta_o = 2.50$). Subsequent paired Wilcoxon test with Bonferroni correction indicated a significant difference between the *lowVR* and the *fullVR* condition ($Z = 2.44$, $p < .05$, 95% CI $[0.50, 2.50]$, $\Theta_{HL} = 1.50$) and the *fullVR* and *embVR* condition ($Z = -2.32$, $p < .05$, 95% CI $[-2.00, -0.00003]$, $\Theta_{HL} = -1.00$), both with medium effect sizes ($r = .46$, 95% CI $[.16, .73]$ and $r = .45$, 95% CI $[.10, .70]$, respectively).

Knowledge Loss for Word-Category Tests: Paired Wilcoxon tests using Bonferroni correction showed no significant differences between post and memory test results for any condition.

5 Discussion

With regards to **H1** insignificant differences between the *lowVR* and the *fullVR* as well as the *lowVR* and the *embVR* conditions imply that increased immersion through an HMD does not necessarily affect associative memory for passive vocabulary. This is aligned with previous results indicating no differences between desktop and fully immersive VR [46]. For embodied interactions Chatain et al. emphasized the importance of the meaning of the interaction (pairing word and object) and the meaning of the concept being learned (affiliation) [4]. While the fully immersive conditions include more bodily action and interaction by moving the objects and words with one's own hands, in *lowVR* embodied interaction is limited to the mouse. Nevertheless, the results imply that increased immersion and the associated differences in embodied interaction do not affect short and long term learning. However, it is another important finding that knowledge loss was significantly higher for the low immersive learning medium, while results of *fullVR* and *embVR* did not significantly differ between post- and memory tests. Increased level of immersion—which is concomitant with a stronger kinesthetic and spatial perception of the words and objects and the possibility to fully focus on the content while naturally interacting with the language through one's own body might impact a person's forgetting curve [13, 41]. This is consistent with previous research, e.g., regarding the benefits of multisensory learning [35, 53], the effects of kinesthetic language learning in VR on learning retention [57], the positive impact of immersive VR in the early acquisition of lexical forms [23], and its positive effects on memorizing foreign-language vocabulary [33].

It is a major finding that for both, short term and long term memorization, increased avatar embodiment and self-representation in the *embVR* condition led to significantly better results compared to the *fullVR* condition. Regarding **H3**, it can be assumed that stronger self-perception through a full-body avatar has an influence on learning. Regarding embodied cognition [29, 54, 56] and the enactive approach to learning [17], a possible explanation is increased self-presence enabling profound embodied interactions through a more holistic and consistent perception of the virtual body interacting with its environment. Considering spontaneous bodily actions while reasoning which play, beside conscious bodily actions, an essential role for modal priming and sensorimotor simulation [4], the absence of a visual representation of spontaneous bodily actions of extremities like the legs in the *fullVR* condition might negatively influence the learning, too. Related to research investigating congruence and realism of avatar embodiment on presence [14, 34, 50, 55, 61] and cognitive load [36], a congruent whole-body representation as a manifestation of the user in the virtual world apparently is an important factor for memorization processes.

The word-category test, like the word-word test, evaluates vocabulary recognition but also measures semantic understanding and abstraction, making it more cognitively demanding. It is therefore not surprising that the performance in both word-word tests was better among all conditions compared to the word-category tests. However, regarding short term memorization it is an important finding that participants significantly performed better in *lowVR* compared to both the *fullVR* and the *embVR* condition. We therefore refuse **H2**. A possible explanation is the immersive nature of the

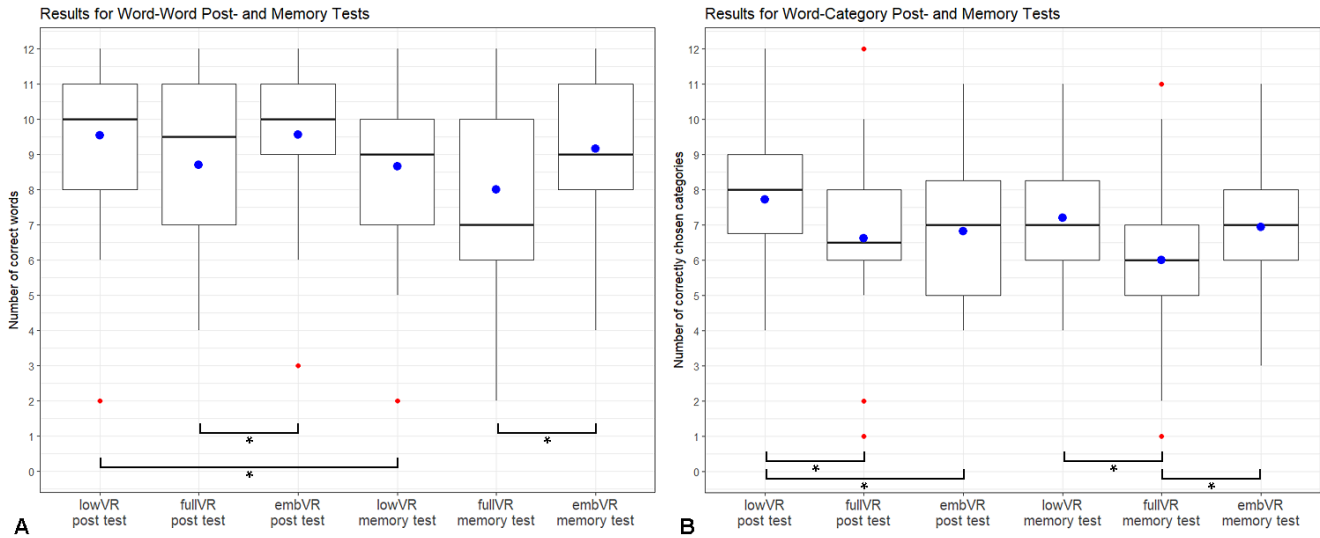


Figure 2: (A) Boxplots results for word-word post- and memory tests. (B) Boxplot results for word-category post- and memory tests. Blue dots represent mean values, outliers are marked as red dots. Significant differences (*) are based on paired Wilcoxon tests with Bonferroni correction.

HMD-based learning environments, providing multi-sensory input and often being associated with a sensation due to technology’s novelty character, may have taken away cognitive resources that could have been used to comprehend the semantic relationship between the words and its hypernyms. Similar distracting effects (e.g., paying attention to details not related to the task) of “new” immersive technologies were reported in previous research [27].

With regards to long term memorization, it is therefore not surprising that participants significantly performed better in the *lowVR* condition of the word-category memory test compared to the *fullVR* condition. However, it is another major finding that participants also performed significantly better in the *embVR* condition compared to the *fullVR* condition while the difference between the *embVR* and *lowVR* condition was insignificant. Again, regarding **H4** an increased avatar embodiment had a positive effect on the learning outcomes and compensated for the different degrees of immersion that were significantly differing in the post-test before. Generally, there was no significant difference between the post and memory tests. However, the results stress the importance of a strong self-perception of bodily interaction and congruent visual representations through a full-body avatar for vocabulary learning in fully immersive VR.

6 Limitations and Future Work

A limitation of the current state of this work is the lack of data on qualitative and self-reported factors, respectively. For example, we did not ask participants regarding perceived cognitive load or task load. Moreover, we do not have information about motivational effects or the general usability. Likewise, it would be interesting to see whether individual language abilities and preconditions in memorization correlate with the study results. With regards to [37], a more granular subdivision of immersion levels such as augmented

reality is of great interest, too. Another parameter that was not assessed are participants’ learning curves. It would be interesting to see whether the pace at which correct word-object pairs are found increases with each iteration. Furthermore, we consider it worth investigating whether there are differences between the different immersion levels and Fitts Law shows additional effects.

As outlined, the short familiarisation phase and a possible novelty effect despite the participants’ above-average previous VR experience could be a factor that influenced the results, too.

While our results highlight the importance of a full-body representation, the effects of morphological divergence between avatars with regards to meaning making should also be considered. This is closely related to research on the Proteus Effect [34, 49].

Eventually, when considering learning content that is more relevant to the target audience, such as French or Spanish vocabulary, using Finnish presents limitations in terms of meaningfulness and motivation. Moreover, tests requiring an active language use, other than the mere recognition of vocabulary, could provide better support.

Future research may include different languages, a larger sample size, a between-participants design, and a mixed-method approach to assess constructs such as usability, task load, flow, and presence.

7 Conclusion

In this study we investigated the difference between low immersive and fully immersive VR as well as the impact of increased avatar embodiment on vocabulary memorization. Specifically, we examined the impact of two different levels of immersion and two clearly distinct degrees of embodiment on passive vocabulary recognition and semantic understanding.

Aligned with previous research, the results imply that fully immersive VR does not improve short- and long-term recognition

compared to desktop VR. However, it prevents knowledge loss, possibly due to multisensory learning. Simultaneously, considering tasks that are more complex than cued recognition, such as abstraction and semantic categorization, fully immersive VR and multisensory input might have led to distraction and higher cognitive load. Regarding embodied learning, our study revealed the positive impact of increased self-perception and embodied interactions. This is an important finding for designers and educators interested in creating fully immersive learning experiences: embodied interaction through full-body avatars can foster long-term vocabulary recognition and semantic understanding.

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A Appendix

Table 1: Exemplary presentation of Word Pool A

				
4 words				
English	rack	cheese	chicken	ax
German	Regal	Käse	Huhn	Axt
Finnish	hylly	juusto	kana	kirves
category	furniture	food	animals	tools
				
4 words				
English	bird	milk	armchair	saw
German	Vogel	Milch	Sessel	Säge
Finnish	lintu	maito	nojatuoli	näki
category	animals	food	furniture	tools
				
4 words				
English	stag	mushroom	cupboard	hammer
German	Hirsch	Pilz	Schrank	Hammer
Finnish	peura	sieni	vaatekappii	vasara
category	animals	food	furniture	tools

Table 2: Word-Word Post Test Results

<i>n</i> = 32	lowVR	fullVR	embVR
mean	9.53	8.69	9.56
SD	2.20	2.48	1.88
median	10.0	9.50	10.0
IQR	3.00	4.00	2.00

Table 3: Word-Word Memory Test Results

<i>n</i> = 32	lowVR	fullVR	embVR
mean	8.66	8.00	9.16
SD	2.35	2.64	1.83
median	9.00	7.00	9.00
IQR	3.00	4.00	2.00

Table 4: Word-Category Post Test Results

<i>n</i> = 32	lowVR	fullVR	embVR
mean	7.72	6.62	6.81
SD	1.87	2.32	2.12
median	8.00	6.50	7.00
IQR	2.25	3.00	3.25

Table 5: Word-Category Memory Test Results

<i>n</i> = 32	lowVR	fullVR	embVR
mean	7.19	6.00	6.94
SD	1.64	2.48	1.76
median	7.00	6.00	7.00
IQR	2.25	2.00	2.00